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SYSTEM STATUS

0.772

TEMPORAL AUC (REALISTIC)

Cross-val AUC: 0.973 (optimistic)

✓ Model loaded

SmartDaaS v1.0
Random Forest - 27,288 records
AUC 0.963 (cross-val)
AUC 0.772 (temporal)

SmartDaaS

AI-Powered HIV Programme Intelligence · Operational Analytics · Decision Support

v1.0 · Decision-Support Platform · Pilot-Ready

Patient Risk Engine

Identify who needs attention most — before they are lost to follow-up.

Upload your patient dataset and get instant risk scores for poor outcomes — adherence issues, treatment interruption, or clinical deterioration. Each score comes with a full SHAP explanation so your team knows exactly why a patient is flagged.

DATA GOVERNANCE

✓ Data governance acknowledged for this session. You may upload patient data.

UPLOAD PATIENT DATA

Upload your patient CSV



Upload

50MB per file · CSV



Use demo data instead



Demo mode: 200 patients from the training set. Upload your own CSV above to analyse real patients.



Using baseline model (Nigerian discovery cohort, AUC: 0.772). Run Local Validation to calibrate for your programme's data.

RISK SUMMARY

70

HIGH RISK ≥70%

23

MEDIUM RISK 40-69%

107

LOW RISK <40%

35%

HIGH RISK RATE



QUICK SHAP PREVIEW — TOP HIGH RISK PATIENT

Why is the highest-risk patient flagged? No need to navigate away.

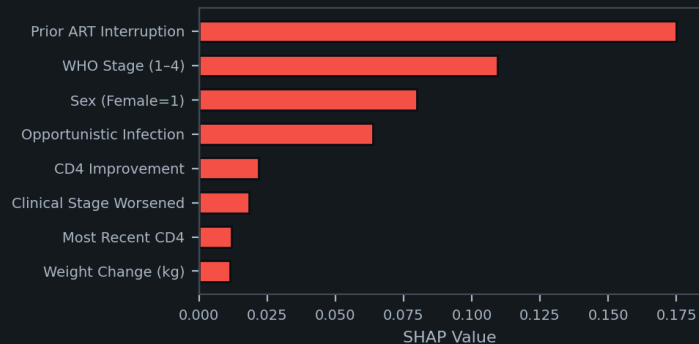
98.9%

PT-0141 · HIGH RISK

Top 3 drivers:

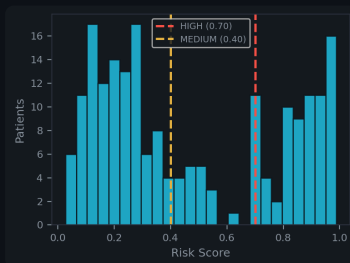
- Prior ART Interruption: ↑ increases risk
- WHO Stage (1-4): ↑ increases risk
- Sex (Female=1): ↑ increases risk

Why PT-0141 is HIGH risk



💡 Go to 🧠 SHAP Explainability for full per-patient waterfall charts across all 15 features for any patient in your cohort.

RISK DISTRIBUTION

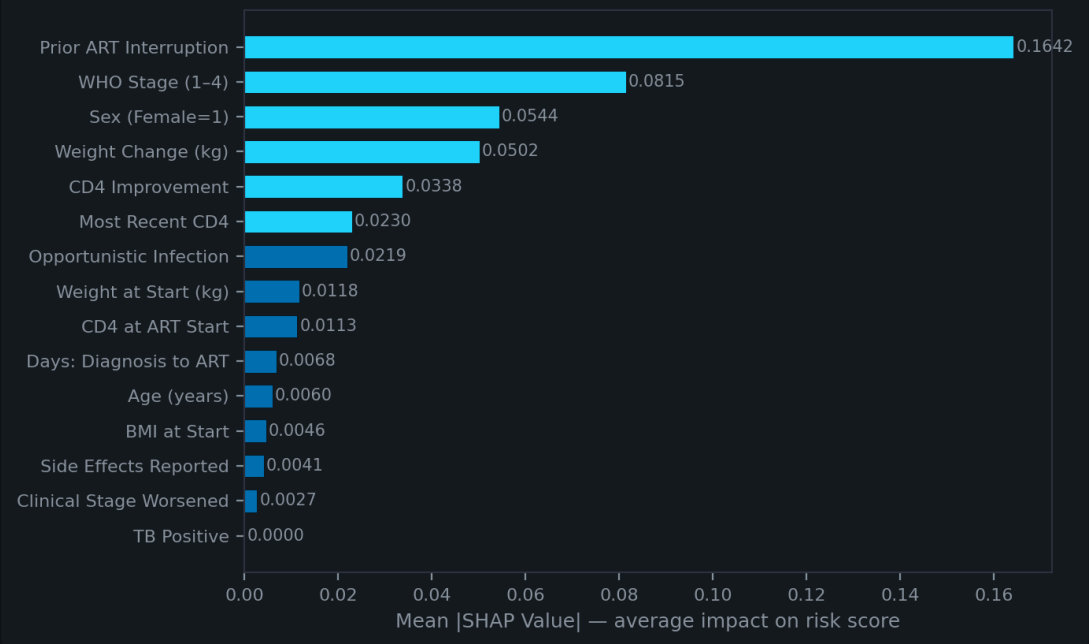


PATIENT RISK SCORES

	patient_id	risk_pct	risk_label	Age	Cd4AtStart	stage_start_num	
141	PT-0141	98.9	HIGH	27.6092	267	1.2318	
29	PT-0029	98.6	HIGH	61.0476	264.1958	1.5608	
69	PT-0069	98.5	HIGH	5.5995	299.3955	1.3004	
84	PT-0084	98.3	HIGH	44.6857	267	3.0393	
177	PT-0177	98.3	HIGH	12.4629	267	1.4629	
108	PT-0108	98	HIGH	41.2418	267	3	

GLOBAL FEATURE IMPORTANCE (SHAP)

Global Feature Importance — All Patients



INDIVIDUAL PATIENT EXPLORER

Select patient (sorted by highest risk):

PT-0141

▼

98.9%

HIGH RISK

Key Clinical Values:

- Age (years): 27.6
- CD4 at ART Start: 267.0
- Most Recent CD4: 396.0
- WHO Stage (1-4): 1.2
- Days: Diagnosis to ART: 0.0
- Prior ART Interruption: 0.8
- TB Positive: 0.0
- Side Effects Reported: 0.0

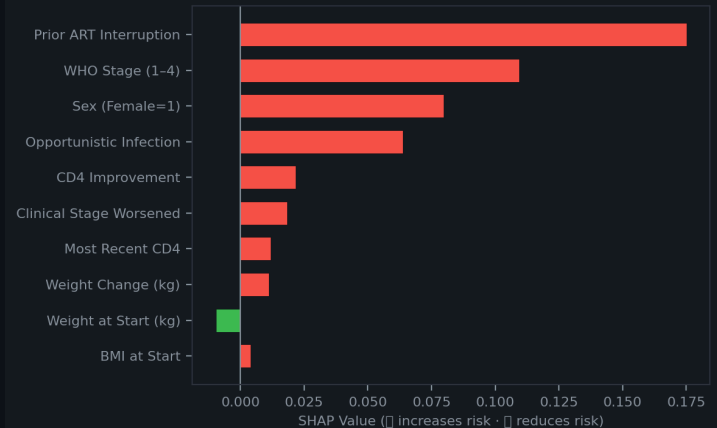
Recommended Actions:

- Schedule urgent adherence counselling within 48 hours
- Activate peer navigator support
- Review regimen tolerability and side effects
- Arrange pill count / home visit
- Escalate to clinical officer for review

SHAP Interpretation Note: Feature contributions reflect model-learned patterns from training data. Individual values should be interpreted alongside clinical judgement and local programme context. Feature contributions may vary across facility types and patient populations. Sex-based contributions reflect cohort-level statistical patterns — not individual clinical determinism. If facility tier data is available, interpret contributions in that structural context.

Local validation is required before deployment within real-world programme environments.

Clinical Feature Contributions: PT-0141 | Risk: 98.9% (HIGH)



EXPORT RESULTS

 All Risk Scores (CSV)

 High Risk Only (70 patients)

 Action List (93 patients)